

1. Explain “Derivative” (or “Differential”).

2. If $f(x) = 5^x$, show that

$$\frac{f(x+h) - f(x)}{h} = 5^x \left(\frac{5^h - 1}{h} \right)$$

3. Find the vertical asymptotes of the function

$$y = \frac{x^2 + 1}{3x - 2x^2}$$

4. A particle moves along a straight line with displacement $s(t)$, velocity $v(t)$, and acceleration $a(t)$. Show that

$$a(t) = v(t) \frac{dv}{ds}$$

Explain the difference between the meanings of the derivatives dv/dt and dv/ds .

5. Prove the mean value theorem.

THE MEAN VALUE THEOREM Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

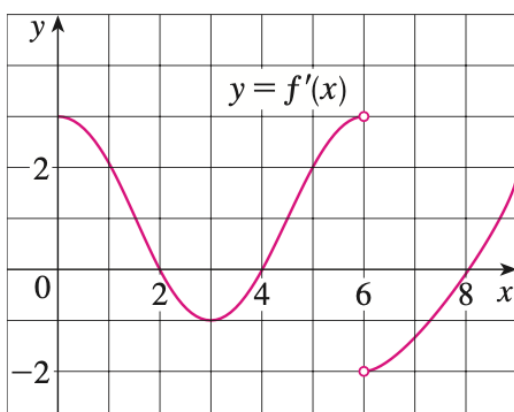
Then there is a number c in (a, b) such that

$$\boxed{1} \quad f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$\boxed{2} \quad f(b) - f(a) = f'(c)(b - a)$$

6. The graph of the derivative f' of a continuous function f is shown as below.



(a) On what intervals is f increasing or decreasing?

(b) At what values of x does f have a local maximum or minimum?

(c) On what intervals is f concave upward or downward?